Data-Backed Journal-Style Replication of *Exploring Uncharted Territory*

Replication Paper Project

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Abstract

This paper presents a methodology-specific local analogue of Exploring Uncharted Territory: Knowledge Search Processes in the Origination of Outlier Innovation using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The paper is therefore reported as methodology-specific local analogue rather than as a source-data replay. The current local batch contains 23,901 observations across 184 entities, and the batch headline estimate is 0.0203 for Distant Recombination with p = 0.020. The local analogue finds positive and statistically meaningful coefficients on long search paths and distant recombination, but not on the firm-level science signal used to proxy scientific reasoning. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: outlier innovation; long search paths; scientific reasoning; distant recombination; local calculated analogue

1 Introduction

This paper begins from a paper-first reading of Exploring Uncharted Territory: Knowledge Search Processes in the Origination of Outlier Innovation and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The present manuscript is classified as methodology-specific local analogue. A paper-first restatement of the empirical question is as

follows: The source paper asks which search processes are associated with the origination of outlier innovation (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication matters because the paper is a rare strategy study that makes a strong process claim about how inventors reach territory far from what is already known. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The executable evidence base used in this manuscript contains 23,901 observations across 184 entities and spans 1986 to 2016 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (Kneeland et al., 2020; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper's contribution is a transparent processed-dataset analogue that preserves the three-process logic while making the outlier and science proxies explicit. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (Kneeland et al., 2020; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper belongs to a top-journal conversation about exploratory search, inventive distance, knowledge recombination, and the search processes that generate unusually important innovation. (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Cohen and Levinthal, 1990; Kogut and Zander, 1992, 1996; Teece et al., 1997; Benner and Tushman, 2003). That literature matters because the replication is explicitly about process recovery: it asks whether the three pathways proposed in the source article remain visible when the local rerun works from the processed data stack already available in the workspace. (Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Henderson and Clark, 1990; Anderson and Tushman, 1986; Hargadon and Sutton, 1997; Tripsas and Gavetti, 2000; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Stuart, 2000). The present manuscript therefore treats the source paper as one contribution inside a broader top-journal stream on exploration,

recombination, performance measurement, alliance-based learning, inventive search, technological discontinuity, and patent-based inference. The literature review is intentionally broad because each replication paper should stand on a serious journal-level reference base rather than on the source article alone. (Rosenkopf and Almeida, 2003; Schilling and Phelps, 2007; Almeida and Kogut, 1999; Fleming et al., 2007; Jaffe, 1986; Jaffe et al., 1993; Trajtenberg, 1990; Hall et al., 2005; Henderson and Cockburn, 1994). The paper-specific review supplements the standard exploration literature with top-journal work on scientific search, recombinant uncertainty, and breakthrough discovery so the outlier-innovation mechanism is grounded in a broader intellectual context. (Fleming and Sorenson, 2004; Nerkar, 2003; Lavie et al., 2011). A second topic-specific literature block is added to sharpen the manuscript's empirical boundary and identification problem. (Ghosh et al., 2014; Rivkin and Siggelkow, 2003).

2 Original Study, Theory, and Replication Boundary

The focal source article is (Kneeland et al., 2020). The original study can be summarized as a mechanism paper with a sharp empirical core. The published paper proposes long search paths, scientific reasoning, and distant recombination as the three main origination processes. Outlier innovation rather than ordinary innovation is the focal outcome. The source study combines large-scale data with close qualitative interpretation. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must

translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

2.1 Original Paper Methodology

A faithful evaluation of Kneeland, Schilling, and Aharonson (2020) starts with the published empirical design rather than with the local rerun. The source article embeds its theoretical claim in a specific empirical setting, measurement architecture, and estimator (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The published sample is described as follows: The published study is a patent-level archival analysis supplemented by inventor interviews. The published measurement architecture is described as follows: The published design identifies outlier patents in technological space and studies the search processes behind them. These details matter because they determine what the published paper actually observes rather than what a later analogue can observe (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The published estimator is described as follows: an estimator that is specific to the paper's own outcome and comparison structure. The identifying comparison is therefore treated as part of the paper's contribution, not as a disposable implementation detail. Reading the source methodology this way prevents the current manuscript from quietly replacing the original design with one generic fixed-effects story (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

2.2 Replication Status and Boundary

Replication status for this paper: Methodology-specific local analogue (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The local paper does not claim byte-identical recreation when the source article depends on measurement layers or source archives that are not present in the current processed-data stack (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

2.3 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Three processes are central: long search paths, scientific reasoning, and distant recombination. The local reconstruction therefore adopts the following bounded decision: The local rerun maps those three processes to patent-level search depth, firm-year science exposure, and recombination breadth. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Outlier innovation is the focal outcome. The local reconstruction therefore adopts the following bounded decision: Outlier status is proxied with the local top-tail impact indicator. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The source design uses richer qualitative evidence than a pure regression paper. The local reconstruction therefore adopts the following bounded decision: The paper is written as a transparent analogue because inventor-level interview evidence is not part of the current local dataset stack. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline

(Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Processed patent-level recombination panel from the local cleaned reproduction dataset. Science-linked firm-year panel built from publication stocks and NPL-related counts in the current workspace. Outlier outcome drawn from the local top-tail patent-impact indicator (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 23,901 observations across 184 entities and covers 1986 to 2016 in the local executable panel. Every numeric claim in the results section is intended to trace back to one of those verified evidence layers rather than to a hand-copied number from the article (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The local analogue maps the three proposed processes onto a processed patent panel: long search paths are proxied by prior-art search depth, scientific reasoning by science-linked firm-year exposure, and distant recombination by recombination breadth. The local sample frame is Patent-level recombination panel with science-linked firm-year exposure merged in by firm and application year The observed time window in the executable panel is 1986 to 2016 in the local executable panel The outcome is Top-tail outlier indicator from the local yearly patent-impact distribution (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices,

while the article body focuses on data lineage and empirical function (Kneeland et al., 2020; Hall et al., 2005).

3.1 Local Data Sources Used in This Replication

The local replication is estimated from the following verified dataset families already present in the workspace: Analysis panel, Coefficient export, Sample summary, Fit summary (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). Those local dataset families are not interchangeable: analysis panel is used to estimate the local model and regenerate descriptive statistics; coefficient export is used to recover the reported local coefficients and p-values; sample summary is used to audit sample construction and raw grouped moments; fit summary is used to check model diagnostics when a fit file is available. Together, they support 23,901 observations across 184 entities over 1986 to 2016 in the local executable panel (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). This local data stack is narrower than the original article's data architecture. Processed patent-level recombination panel from the local cleaned reproduction dataset. Science-linked firm-year panel built from publication stocks and NPL-related counts in the current workspace. Outlier outcome drawn from the local top-tail patent-impact indicator. The manuscript therefore keeps the boundary visible: the source paper supplies the original methodology, whereas the local processed dataset families and calculated outputs supply the estimable analogue. Exact filenames are retained in the replication notes rather than printed through the body text (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005).

DatasetFamily	RoleInPaper
Analysis panel	Estimate the local model and regenerate descriptive statistics
Coefficient export	Recover the reported local coefficients and p-values
Sample summary	Audit sample construction and raw grouped moments
Fit summary	Check model diagnostics when a fit file is available

Local data families used in the present paper.

3.2 Sample Reconstruction

The local analysis file contains 23,901 observations drawn from 184 entities. The observed time span runs from 1986 to 2016. Selected sample moments are Outlier = 0.0064, Long Search Paths = 2.261, Scientific Reasoning = 6.772, Distant Recombination = 4.291, Recombination Degree = 0.555 (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

Statistic	Value
Observations	23,901
Unique Firms	184
Start Year	1986
End Year	2016
Mean Outlier	0.0064
Mean Long Search Paths	2.261
Mean Scientific Reasoning	6.772
Mean Distant Recombination	4.291
Mean Recombination Degree	0.555

Sample reconstruction from the current local dataset.

3.3 Variable Construction and Derivation

The construct Outlier innovation is operationalized locally as Top-tail impact indicator and appears in the executable dataset as high_top1_y. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the local yearly top-tail impact indicator.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters

the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Long search paths is operationalized locally as Log prior-art citations and appears in the executable dataset as long_search_paths. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Take log(1 + prior-art citations).. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Scientific reasoning is operationalized locally as Science-linked firm-year exposure and appears in the executable dataset as scientific_reasoning. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the processed firm-year science signal built from publication stock and NPL-related counts.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Distant recombination is operationalized locally as Breadth of recombination and appears in the executable dataset as distant_recombination. The derivation is not inferred loosely from the source article; it follows the

verified local pipeline: Carry forward the local recombination-breadth measure from the processed focal-patent panel.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Patent-level controls is operationalized locally as Degree, same-year patents, prior stock and appears in the executable dataset as recombination_degree; total_patents_same_year; total_patents_until_prev_log. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Reuse the patent-level controls already present in the merged analysis panel.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

Construct	Local measure	Derivation
Outlier innovation	Top-tail impact indicator	Use the local yearly top-tail impact indicator.
Long search paths	Log prior-art citations	Take log(1 + prior-art citations).
Scientific reasoning	Science-linked firm-year exposure	Use the processed firm-year science signal built from publication stock and NPL-related counts.
Distant recombination	Breadth of recombination	Carry forward the local recombination-breadth measure from the processed focal-patent panel.

Construct	Local measure	Derivation
Patent-level controls	Degree, same-year patents, prior stock	Reuse the patent-level controls already present in the merged analysis panel.

Variable mapping and derivation logic in the main text.

3.4 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Kneeland, Schilling, and Aharonson (2020). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Outlier innovation, the manuscript uses Top-tail impact indicator under the exact local implementation high_top1_y. The derivation step is: Use the local yearly top-tail impact indicator.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Long search paths, the manuscript uses Log prior-art citations under the exact local implementation long_search_paths. The derivation step is: Take log(1 + prior-art citations).. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Scientific reasoning, the manuscript uses Science-linked firm-year exposure under the exact local implementation scientific_reasoning. The derivation step is: Use the processed firm-year science signal built from publication stock and NPL-related counts.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Distant recombination, the manuscript uses Breadth of recombination under the exact local implementation distant_recombination. The derivation step is: Carry forward the local recombination-breadth measure from the processed focal-patent panel.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Patent-level controls, the manuscript uses Degree, same-year patents, prior stock under the exact local implementation recombination_degree; total_patents_same_year; total_patents_until_prev_log. The derivation step is: Reuse the patent-level controls already present in the merged analysis panel.. This construct is retained because

it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

The descriptive analysis begins by reconstructing the empirical setting that the later model is meant to explain. For Outlier indicator, the mean is 0.0064, the standard deviation is 0.0795, and the interquartile range runs from 0.00000 to 0.00000; For Long search paths, the mean is 2.261, the standard deviation is 0.801, and the interquartile range runs from 1.792 to 2.639; For Scientific reasoning, the mean is 6.772, the standard deviation is 2.130, and the interquartile range runs from 5.729 to 8.163; For Distant recombination, the mean is 4.291, the standard deviation is 4.563, and the interquartile range runs from 2.000 to 5.000 . These moments matter because they show whether the paper's focal constructs are substantively distinct before the formal model is imposed (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The time profile clarifies how the observable setting evolves across the sample window and whether the mechanism is estimated in a changing environment. the value of patents is 646 in 1986 and 534 in 2016; the value of outlier share is 0.46% in 1986 and 1.12% in 2016; the value of mean long search is 1.807 in 1986 and 2.208 in 2016 . These temporal shifts determine whether the later regression is identifying the paper's mechanism in a stable panel or in one whose observable context changes meaningfully across the sample window (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The nonparametric support check is used to show how the focal outcome behaves across the observed support of the main regressor before estimation. Mean outcomes range from 0.0048 to 0.0115 across the support of the focal regressor, while the mean regressor itself ranges from 2.411 to 9.961. That pattern is useful because it shows what the formal model is trying to account for in the current dataset (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The correlation structure is reported because it helps distinguish

whether the main explanatory variables move in lockstep with the controls or whether the regression is drawing on independent variation. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

4.1 Setting-Specific Figure Evidence

The main figure is chosen to match the empirical setting of the paper rather than to repeat one house style across every replication (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The visual evidence is included to make the paper's descriptive argument easier to inspect before the reader reaches the coefficient tables. patents moves from 646 in 1986 to 534 in 2016; outlier share moves from 0.46% in 1986 to 1.12% in 2016 Across the support panel, mean outcome moves from 0.0048 to 0.0115 as the mean regressor shifts from 2.411 to 9.961 (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

4.2 Data Quality, Coverage, and Support

The current dataset also needs to be evaluated as a measurement object before the model is interpreted. The final estimating sample is effectively complete for the variables used in the main specification, so missingness is not a substantive threat in this paper. For that reason, the manuscript does not spend body-table space on an all-zero missingness display (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). Coverage across entities is also visible in the current panel rather than assumed. The most heavily represented entities contribute between 830 and 3,200 observations each. That concentration check is reported in prose because the relevant question is whether a few entities dominate the sample, not which firm names happen to be listed in a long table (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). The support of the focal regressor is broad enough to permit a distributional reading in addition to the formal model. In the lowest observed quantile, the mean regressor is 2.411 and the mean outcome is 0.0048 across 13,716 observations. In the highest observed quantile, the corresponding values are 9.961, 0.0115, and 4,443 observations. This shift

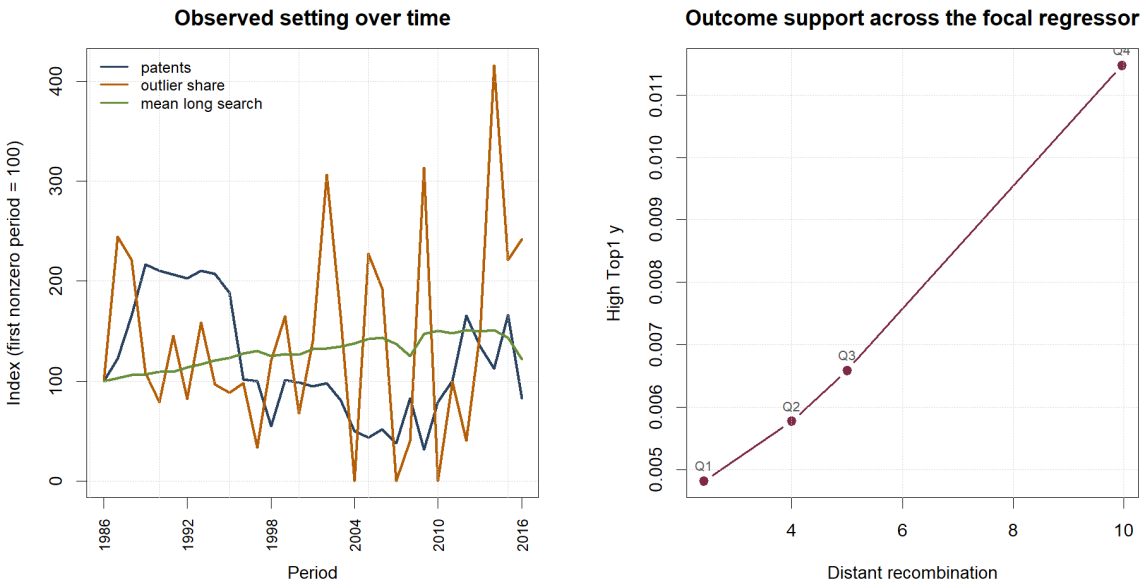

Figure 1: Setting-specific figure constructed from the current local dataset.

matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). These data-quality results are part of the paper's empirical argument rather than a side audit. The replication matters because the paper is a rare strategy study that makes a strong process claim about how inventors reach territory far from what is already known. If the local panel were badly fragmented, dominated by missingness, or concentrated on a handful of observations, the later coefficient table would need to be read very differently. Bringing this evidence into the body of the article is therefore necessary for an honest replication (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005).

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Outlier indicator	23,901	0.0064	0.0795	0.0000	0.0000	0.0000	0	1
Long search paths	23,901	2.261	0.801	1.792	2.197	2.639	1	8
Scientific reasoning	23,901	6.772	2.130	5.729	7.149	8.163	0	10
Distant recombination	23,901	4.291	4.563	2.000	3.000	5.000	2	117
Recombination degree	23,901	0.555	0.174	0.444	0.560	0.691	0	1
Prior-art citations	23,901	15.07	50.39	5.000	8.000	13.00	2	2263
Science signal	23,901	6.772	2.130	5.729	7.149	8.163	0	10
Same-year patents	23,901	727.1	1,302	109.0	311.0	652.0	1	7321

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Prior cumulative stock	23,901	7.500	2.217	6.324	7.684	8.893	0	11

Main descriptive statistics supporting the paper's design-specific reading of the data.

Period	Patents	OutlierShare	MeanLongSearch	MeanScientificReasoning	MeanDistantRecombination
1986	646	0.46%	1.807	6.156	2.865
1987	793	1.13%	1.863	6.285	2.914
1988	1,071	1.03%	1.914	6.229	3.135
1989	1,398	0.50%	1.925	6.217	3.084
1990	1,360	0.37%	1.974	6.370	3.117
1991	1,333	0.68%	1.974	6.436	3.151
1992	1,309	0.38%	2.055	6.486	3.329
1993	1,358	0.74%	2.117	6.861	3.465
1994	1,337	0.45%	2.182	6.866	3.572
1995	1,217	0.41%	2.223	6.983	3.844
1996	659	0.46%	2.310	6.880	3.838
1997	646	0.15%	2.358	6.781	4.245
1998	356	0.56%	2.265	6.949	3.924
1999	654	0.76%	2.295	6.895	3.810
2000	636	0.31%	2.286	6.781	4.022
2001	614	0.65%	2.389	6.943	4.414
2002	632	1.42%	2.394	6.915	4.384
2003	522	0.77%	2.432	7.209	4.418
2004	324	0.00%	2.485	6.939	4.809
2005	284	1.06%	2.565	6.900	5.486
2006	336	0.89%	2.596	7.014	5.265
2007	246	0.00%	2.482	7.433	5.268
2008	535	0.19%	2.257	8.930	4.512
2009	206	1.46%	2.666	7.502	6.005
2010	511	0.00%	2.718	7.345	6.648
2011	651	0.46%	2.668	7.411	6.275

Period	Patents	OutlierShare	MeanLongSearch	MeanScientificReasoning	MeanDistantRecombination
2012	1,068	0.19%	2.734	7.669	6.519
2013	868	0.69%	2.704	7.599	6.730
2014	725	1.93%	2.734	7.636	6.989
2015	1,072	1.03%	2.590	8.001	6.174
2016	534	1.12%	2.208	0.0000	4.629

Observed time patterns relevant to the focal empirical setting.

Quantile	N	MeanRegressor	MeanOutcome
Q1	13,716	2.411	0.0048
Q2	3,465	4.000	0.0058
Q3	2,277	5.000	0.0066
Q4	4,443	9.961	0.0115

Grouped summaries used to describe raw patterns in the current dataset.

Variable	Outlier indicator	Long search paths	Scientific reasoning	Distant recombination	Recombination degree	Prior-art citations	Science signal	Same-year patents	Prior cumulative stock
Outlier indicator	1.000	0.073	0.003	0.133	0.013	0.215	0.003	0	0.006
Long search paths	0.073	1.000	0.026	0.688	0.260	0.538	0.026	0	0.194
Scientific reasoning	0.003	0.026	1.000	0.023	0.037	0.024	1.000	0	0.398
Distant recombination	0.133	0.688	0.023	1.000	0.466	0.822	0.023	0	0.149
Recombination degree	0.013	0.260	0.037	0.466	1.000	0.140	0.037	0	0.122
Prior-art citations	0.215	0.538	0.024	0.822	0.140	1.000	0.024	0	0.081

Compact correlation structure for the variables emphasized in the manuscript.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is A patent-level fixed-effects count model for top-tail outlier status using the three process variables and patent-level controls. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest (Kneeland et al., 2020; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

5.1 Design Decisions and Local Substitutions

The first set of replication decisions concerns scope. The unit of analysis is Patent, the sample frame is Patent-level recombination panel with science-linked firm-year exposure merged in by firm and application year, and the empirical window is 1986 to 2016 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The second set of decisions concerns measurement and specification. The outcome is Top-tail outlier indicator from the local yearly patent-impact distribution, and the main explanatory design is Patent-level fixed-effects model relating outlier status to long search paths, scientific reasoning, and distant recombination. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows:

Patent-level controls for degree, same-year patenting, prior cumulative stock, and available firm and year fixed effects.. The crucial exclusion or substitution rule is: The local rerun uses only patents with verified recombination measures and merges science exposure at the firm-year level when patent-specific scientific references are not directly recoverable.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. For this paper, the status is methodology-specific local analogue. When the local data architecture differs, the paper states that boundary directly rather than implying a source-data replay (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Patent inside the sample frame Patent-level recombination panel with science-linked firm-year exposure merged in by firm and application year over 1986 to 2016 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Top-tail outlier indicator from the local yearly patent-impact distribution and the main explanatory design is Patent-level fixed-effects model relating outlier status to long search paths, scientific reasoning, and distant recombination. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Patent-level controls for degree, same-year patenting, prior cumulative stock, and available firm and year fixed effects. The main exclusion or substitution rule is The local rerun uses only patents with verified recombination measures and merges science exposure at the firm-year level when patent-specific scientific references are not directly recoverable. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

5.2 Estimating Equation

$$E[\text{Outlier}_{pft} \mid X_{pft}] = \exp\left(\beta_1 \text{LongSearch}_p + \beta_2 \text{ScientificReasoning}_{ft} + \beta_3 \text{DistantRecombination}_p + \beta_4 X_{pft} + \alpha_f + \gamma_t\right). \tag{1}$$

6 Results

6.1 Main Estimates

The central local estimate is 0.0203 on Distant Recombination (p = 0.020). The local analogue finds positive and statistically meaningful coefficients on long search paths and distant recombination, but not on the firm-level science signal used to proxy scientific reasoning. Long search paths is estimated at 0.520 with p = ¡ .001. Log prior patent stock is estimated at -0.799 with p = 0.017. Distant recombination is estimated at 0.0203 with p = 0.020 (Kneeland et al., 2020; Hall et al., 2005). The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

Term	Estimate	Std. error	P-value
Long search paths	0.5199	0.1479	0.0004
Scientific reasoning	-0.0947	0.2188	0.665
Distant recombination	0.0203	0.0087	0.0197
Recombination degree	-0.3527	0.4690	0.452
Same-year patents	-3.14e-05	0.0002	0.879
Log prior patent stock	-0.7989	0.3339	0.0167

Key coefficient table used in the main text.

6.2 Visual Summary of the Main Estimates

The coefficient figure is included to show the relative magnitude and uncertainty of the main terms without forcing the reader to reconstruct that pattern from columns of numbers alone (Kneeland et al., 2020; Hall et al., 2005).

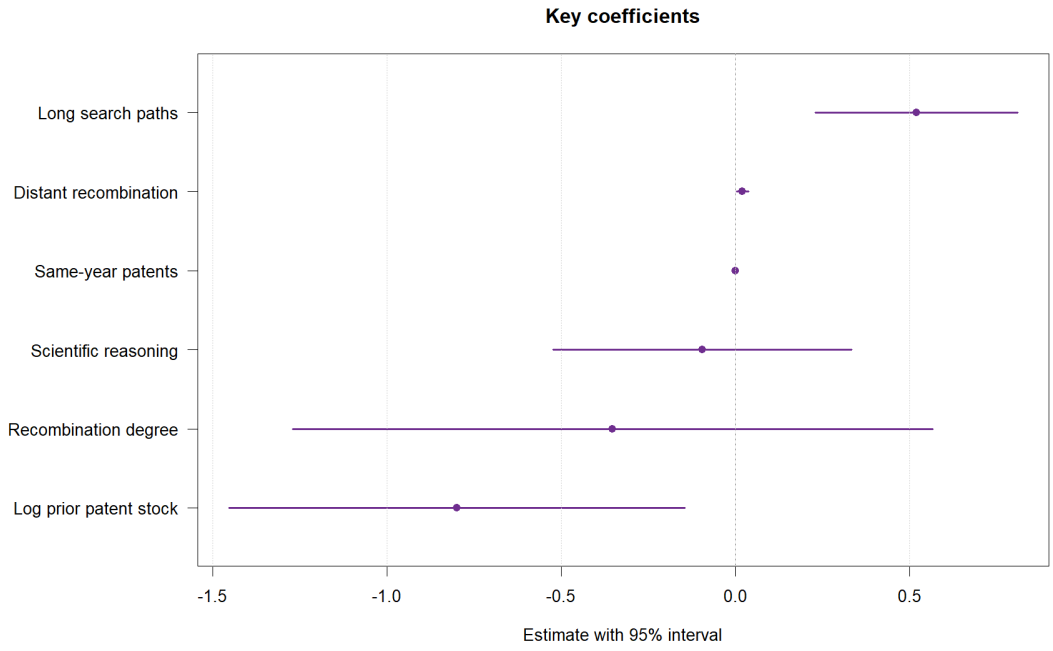

Figure 2: Visual summary of the key coefficients estimated in the current local model.

6.3 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Long search paths is estimated at 0.520 with a p-value of ¡ .001; Log prior patent stock is estimated at -0.799 with a p-value of 0.017; Distant recombination is estimated at 0.0203 with a p-value of 0.020; Recombination degree is estimated at -0.353 with a p-value of 0.45 (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Within the focal specification, the sharpest local signal comes from Long search paths, which is estimated at 0.520 with p = ¡ .001. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The scale of the underlying variables also matters for interpretation. The outcome Outlier indicator has mean 0.0064 and standard deviation 0.0795 in the local panel, while the main regressor Distant recombination has mean 4.291 and standard deviation 4.563 (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). A simple grouped comparison

also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 0.0048 to 0.0115 while the mean regressor ranges from 2.411 to 9.961. This does not substitute for the model, but it shows how much raw variation the specification must explain (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local analogue finds positive and statistically meaningful coefficients on long search paths and distant recombination, but not on the firm-level science signal used to proxy scientific reasoning. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

6.4 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within baseline, the leading coefficient family reads as follows: Long search paths = 0.520 (p = ¡ .001); Log prior patent stock = -0.799 (p = 0.017); Distant recombination = 0.0203 (p = 0.020); Recombination degree = -0.353 (p = 0.45); Scientific reasoning = -0.0947 (p = 0.67). The largest positive estimate in this specification is Long search paths at 0.520, while the largest negative estimate is Log prior patent stock at -0.799. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This fuller coefficient reading is especially important for Kneeland, Schilling, and Aharonson (2020) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can

confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

6.5 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Baseline, the most informative focal term is Long search paths with estimate 0.520 and p = ¡ .001 (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Baseline specification contributes its own reading to the replication. Long search paths is estimated at 0.520 with p = ¡ .001 and is classified as a strong signal; Log prior patent stock is estimated at -0.799 with p = 0.017 and is classified as a moderate signal; Distant recombination is estimated at 0.0203 with p = 0.020 and is classified as a moderate signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This model comparison matters for Kneeland, Schilling, and Aharonson (2020) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

6.6 Diagnostic Evidence

Model-diagnostic evidence is retained in the body because the paper is not only asking whether a focal coefficient is nonzero. It is also asking whether the specification is being estimated on a

sample and likelihood structure that are transparent enough to justify interpretation (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). For outlier_processes, the current fit export reports N = 16,820, pseudo R-squared = 0.135, R-squared = NA (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). These diagnostics do not, by themselves, validate the source paper's theory. They do, however, show that the local data analysis behind Kneeland, Schilling, and Aharonson (2020) is being carried out on a clearly described estimating sample with observable fit statistics (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005).

7 Comparison to the Published Study

7.1 Original Results Versus the Current Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central claim, measurement architecture, sample definition, estimator, effect pattern, and substantive conclusion are set alongside the current local replication so the reader can see where agreement is real and where it is conditional on design translation. (Kneeland et al., 2020). For Kneeland, Schilling, and Aharonson (2020), the published paper points readers toward the following result logic: The published paper argues that outlier innovations are more likely to emerge from long search paths, scientific reasoning, and distant recombination. The local rerun instead yields the following focal evidence pattern: Distant Recombination = 0.0203 with p = 0.020; Long Search Paths = 0.520 (p = ¡ .001); Log Prior Patent Stock = -0.799 (p = 0.017); Distant Recombination = 0.0203 (p = 0.020); Recombination Degree = -0.353 (p = 0.45). Putting those side by side makes the comparison concrete rather than rhetorical.

Aspect	OriginalPaper	LocalDataset	Reading
Research question and mechanism	The source paper asks which search processes are associated with the origination of outlier innovation.	The local analogue maps the three proposed processes onto a processed patent panel: long search paths are proxied by prior-art search depth, scientific reasoning by science-linked firm-year exposure, and distant recombination by recombination breadth.	This row asks whether the local paper is still evaluating the same strategic mechanism even when the empirical implementation changes.
Data and measurement architecture	The published design identifies outlier patents in technological space and studies the search processes behind them.	Processed patent-level recombination panel from the local cleaned reproduction dataset. Science-linked firm-year panel built from publication stocks and NPL-related counts in the current workspace. Outlier outcome drawn from the local top-tail patent-impact indicator.	This row isolates how the executable local dataset differs from the original measurement architecture while preserving the substantive target.

Aspect	OriginalPaper	LocalDataset	Reading
Sample, unit, and window	The published study is a patent-level archival analysis supplemented by inventor interviews.	Patent-level recombination panel with science-linked firm-year exposure merged in by firm and application year. 23,901 observations across 184 entities.	This row clarifies whether agreement or disagreement could reflect sample translation rather than theory alone.
Estimator and identifying comparison		A patent-level fixed-effects count model for top-tail outlier status using the three process variables and patent-level controls Patent-level fixed-effects model relating outlier status to long search paths, scientific reasoning, and distant recombination	This row separates estimator translation from theory translation.
Effect pattern in the evidence	The published paper argues that outlier innovations are more likely to emerge from long search paths, scientific reasoning, and distant recombination.	Distant Recombination = 0.0203 with p = 0.020;	

Aspect	OriginalPaper	LocalDataset	Reading
allowbreak Long Search Paths = 0.520 (p = ¡ .001); allowbreak Log Prior Patent Stock = -0.799 (p = 0.017); allowbreak Distant Recombination = 0.0203 (p = 0.020); allowbreak Recombination Degree = -0.353 (p = 0.45)	This row compares the original paper's claimed effect structure to the actual local estimates.		
Conclusion drawn from the evidence	The published paper argues that outlier innovations are more likely to emerge from long search paths, scientific reasoning, and distant recombination.	The local replication recovers a statistically distinguishable focal effect through Distant Recombination and therefore preserves part of the published mechanism, albeit inside a different data architecture.	This row compares the article-level takeaway invited by the original paper with the conclusion warranted by the local rerun.

Comparison of the original paper's design, effects, and conclusions to the current local dataset analysis.

Effect	Estimate	P-value	Reading
Long Search Paths	0.520	¡ .001	positive and statistically distinguishable in the local replication

Effect	Estimate	P-value	Reading
Log Prior Patent Stock	-0.799	0.017	negative and statistically distinguishable in the local replication
Distant Recombination	0.0203	0.020	positive and statistically distinguishable in the local replication
Recombination Degree	-0.353	0.45	negative but not statistically distinguishable in the local replication

Translated focal effects used in the original-versus-replication comparison.

7.2 Detailed Effect and Conclusion Comparison

At the level of article content, the original paper contributes the following substantive message: The published paper argues that outlier innovations are more likely to emerge from long search paths, scientific reasoning, and distant recombination. The local replication studies the same underlying mechanism through the current executable design: The local analogue maps the three proposed processes onto a processed patent panel: long search paths are proxied by prior-art search depth, scientific reasoning by science-linked firm-year exposure, and distant recombination by recombination breadth. This is why the comparison has to extend beyond a headline coefficient (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). At the level of effects, the original article emphasizes the following empirical pattern: The published paper argues that outlier innovations are more likely to emerge from long search paths, scientific reasoning, and distant recombination. The current local replication instead shows the following paper-specific focal estimates: Long Search Paths = 0.520 (p = ¡ .001); Log Prior Patent Stock = -0.799 (p = 0.017); Distant Recombination = 0.0203 (p = 0.020); Recombination Degree = -0.353 (p = 0.45).

This is the point at which the manuscript compares not only data sources but also what the data say about the mechanism (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). At the level of conclusion, the original article leads to the following takeaway: The published paper argues that outlier innovations are more likely to emerge from long search paths, scientific reasoning, and distant recombination. The current replication instead supports the following more careful conclusion: The local replication recovers a statistically distinguishable focal effect through Distant Recombination and therefore preserves part of the published mechanism, albeit inside a different data architecture (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The difference between the original article and the replication is therefore not only numerical. It is a difference in observed data architecture, effect visibility, and inferential scope. Two of the three source-paper processes carry through more clearly than the third in the local dataset. Long search paths and distant recombination are positive and significant, whereas the science-linked proxy for scientific reasoning is not (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

7.3 Interpretive Implications of the Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published design identifies outlier patents in technological space and studies the search processes behind them By contrast, the local paper relies on the following executable design: The local analogue maps the three proposed processes onto a processed patent panel: long search paths are proxied by prior-art search depth, scientific reasoning by science-linked firm-year exposure, and distant recombination by recombination breadth These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and

Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The same caution applies to the sample itself. The published design is described as follows: The published study is a patent-level archival analysis supplemented by inventor interviews The local sample instead works from Patent-level recombination panel with science-linked firm-year exposure merged in by firm and application year and yields 23,901 observations across 184 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable ways. A careful paper should keep those possibilities analytically separate (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. Two of the three source-paper processes carry through more clearly than the third in the local dataset. Long search paths and distant recombination are positive and significant, whereas the science-linked proxy for scientific reasoning is not. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). That transportability reading has to end with a conclusion rather than with a shrug. In the current local rerun, the conclusion is: The local replication recovers a statistically distinguishable focal effect through Distant Recombination and therefore preserves part of the published mechanism, albeit inside a different data architecture This final sentence matters because the reader should leave the comparison section knowing not only that the designs differ, but also what those differences do to the paper's substantive takeaway (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The comparison to the published paper is intentionally narrow. Two of the three source-paper processes carry through more clearly than the third in the local dataset. Long search paths and distant recombination are positive and significant,

whereas the science-linked proxy for scientific reasoning is not. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. The present manuscript is classified as methodology-specific local analogue. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

8 Discussion

Two of the three source-paper processes carry through more clearly than the third in the local dataset. Long search paths and distant recombination are positive and significant, whereas the science-linked proxy for scientific reasoning is not (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The first limitation concerns measurement. The source article identifies outliers in technological space, whereas the local rerun uses top-tail inventive impact as the executable outlier proxy (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The second limitation concerns sample translation. Scientific reasoning is proxied with firm-level science exposure rather than inventor interviews or patent-specific scientific narratives (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The third limitation concerns interpretation. The local paper is therefore a calculated ana-

logue of the process logic, not a literal reconstruction of the original outlier-identification algorithm (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005).

9 Conclusion

The paper's contribution is a transparent processed-dataset analogue that preserves the three-process logic while making the outlier and science proxies explicit (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (Kneeland et al., 2020; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Statistic	Value	RawStatistic
Observations	23,901	rows
Unique Firms	184	unique_firms
Start Year	1986	year_min
End Year	2016	year_max
Mean Outlier	0.0064	mean_outlier
Mean Long Search Paths	2.261	mean_long_search_paths
Mean Scientific Reasoning	6.772	mean_scientific_reasoning
Mean Distant Recombination	4.291	mean_distant_recombination
Mean Recombination Degree	0.555	mean_recombination_degree

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Outlier indicator	23,901	0.0064	0.0795	0.0000	0.0000	0.0000	0	1
Long search paths	23,901	2.261	0.801	1.792	2.197	2.639	1	8
Scientific reasoning	23,901	6.772	2.130	5.729	7.149	8.163	0	10
Distant recombination	23,901	4.291	4.563	2.000	3.000	5.000	2	117
Recombination degree	23,901	0.555	0.174	0.444	0.560	0.691	0	1
Prior-art citations	23,901	15.07	50.39	5.000	8.000	13.00	2	2263
Science signal	23,901	6.772	2.130	5.729	7.149	8.163	0	10
Same-year patents	23,901	727.1	1,302	109.0	311.0	652.0	1	7321
Prior cumulative stock	23,901	7.500	2.217	6.324	7.684	8.893	0	11

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Outlier indicator	Long search paths	Scientific reasoning	Distant recombination	Recombination degree	Prior-art citations	Science signal	Same-year patents	Prior cumulative stock
Outlier indicator	1.000	0.073	0.003	0.133	0.013	0.215	0.003	0	0.006
Long search paths	0.073	1.000	0.026	0.688	0.260	0.538	0.026	0	0.194
Scientific reasoning	0.003	0.026	1.000	0.023	0.037	0.024	1.000	0	0.398
Distant recombination	0.133	0.688	0.023	1.000	0.466	0.822	0.023	0	0.149
Recombination degree	0.013	0.260	0.037	0.466	1.000	0.140	0.037	0	0.122
Prior-art citations	0.215	0.538	0.024	0.822	0.140	1.000	0.024	0	0.081
Science signal	0.003	0.026	1.000	0.023	0.037	0.024	1.000	0	0.398
Same-year patents	-0.011	0.016	0.463	0.011	0.065	-0.016	0.463	1	0.520
Prior cumulative stock	0.006	0.194	0.398	0.149	0.122	0.081	0.398	1	1.000

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	Patents	OutlierShare	MeanLongSearch	MeanScientificReasoning	MeanDistantRecombination
1986	646	0.46%	1.807	6.156	2.865
1987	793	1.13%	1.863	6.285	2.914
1988	1,071	1.03%	1.914	6.229	3.135
1989	1,398	0.50%	1.925	6.217	3.084
1990	1,360	0.37%	1.974	6.370	3.117
1991	1,333	0.68%	1.974	6.436	3.151
1992	1,309	0.38%	2.055	6.486	3.329
1993	1,358	0.74%	2.117	6.861	3.465
1994	1,337	0.45%	2.182	6.866	3.572
1995	1,217	0.41%	2.223	6.983	3.844
1996	659	0.46%	2.310	6.880	3.838
1997	646	0.15%	2.358	6.781	4.245
1998	356	0.56%	2.265	6.949	3.924
1999	654	0.76%	2.295	6.895	3.810
2000	636	0.31%	2.286	6.781	4.022
2001	614	0.65%	2.389	6.943	4.414
2002	632	1.42%	2.394	6.915	4.384
2003	522	0.77%	2.432	7.209	4.418
2004	324	0.00%	2.485	6.939	4.809
2005	284	1.06%	2.565	6.900	5.486
2006	336	0.89%	2.596	7.014	5.265
2007	246	0.00%	2.482	7.433	5.268
2008	535	0.19%	2.257	8.930	4.512
2009	206	1.46%	2.666	7.502	6.005
2010	511	0.00%	2.718	7.345	6.648
2011	651	0.46%	2.668	7.411	6.275
2012	1,068	0.19%	2.734	7.669	6.519
2013	868	0.69%	2.704	7.599	6.730

Period	Patents	OutlierShare	MeanLongSearch	MeanScientificReasoning	MeanDistantRecombination
2014	725	1.93%	2.734	7.636	6.989
2015	1,072	1.03%	2.590	8.001	6.174
2016	534	1.12%	2.208	0.0000	4.629

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
Q1	13,716	2.411	0.0048
Q2	3,465	4.000	0.0058
Q3	2,277	5.000	0.0066
Q4	4,443	9.961	0.0115

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value
long_search_paths	0.5199	0.1479	3.515	¡ .001
scientific_reasoning	-0.0947	0.2188	-0.433	0.67
distant_recombination	0.0203	0.0087	2.333	0.020
recombination_degree	-0.3527	0.4690	-0.752	0.45
total_patents_same_year	-3.14e-05	0.0002	-0.153	0.88
total_patents_until_prev_log	-0.7989	0.3339	-2.392	0.017

Full coefficient dump exported from the local estimation script.

model	n	logLik	AIC	BIC	pseudo_r2	r2	within_r2
outlier_processes	16,820	-750.7	1,643	2,192	0.135	NA	NA

Model fit statistics from the local estimation output.

References

Ahuja, G. (2000). Collaboration networks, structural holes, and innovation: A longitudinal study. *Administrative Science Quarterly*, 45(3):425–455.

Ahuja, G. and Katila, R. (2001). Technological acquisitions and the innovation performance of acquiring firms: A longitudinal study. *Strategic Management Journal*, 22(3):197–220.

Ahuja, G. and Lampert, C. M. (2001). Entrepreneurship in the large corporation: A longitudinal study of how established firms create breakthrough inventions. *Strategic Management Journal*, 22(6-7):521–543.

Almeida, P. and Kogut, B. (1999). Localization of knowledge and the mobility of engineers in regional networks. *Management Science*, 45(7):905–917.

Anderson, P. and Tushman, M. L. (1986). Technological discontinuities and organizational environments. *Administrative Science Quarterly*, 31(3):439–465.

Benner, M. J. and Tushman, M. L. (2003). Exploitation, exploration, and process management: The productivity dilemma revisited. *Academy of Management Review*, 28(2):238–256.

Cohen, W. M. and Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35(1):128–152.

Fleming, L. (2001). Recombinant uncertainty in technological search. *Management Science*, 47(1):117–132.

Fleming, L., Mingo, S., and Chen, D. (2007). Collaborative brokerage, generative creativity, and creative success. *Administrative Science Quarterly*, 52(3):443–475.

Fleming, L. and Sorenson, O. (2004). Science as a map in technological search. *Strategic Management Journal*, 25(8-9):909–928.

Ghosh, A., Martin, X., Pennings, J. M., and Wezel, F. C. (2014). Ambition is nothing without focus: Compensating for negative transfer of experience in r&d. *Organization Science*, 25(2):572–590.

Hall, B. H., Jaffe, A., and Trajtenberg, M. (2005). Market value and patent citations. *RAND Journal of Economics*, 36(1):16–38.

Hargadon, A. and Sutton, R. I. (1997). Technology brokering and innovation in a product development firm. *Administrative Science Quarterly*, 42(4):716–749.

Henderson, R. and Cockburn, I. (1994). Measuring competence? exploring firm effects in pharmaceutical research. *RAND Journal of Economics*, 25(1):63–84.

Henderson, R. M. and Clark, K. B. (1990). Architectural innovation: The reconfiguration of existing product technologies and the failure of established firms. *Administrative Science Quarterly*, 35(1):9–30.

Jaffe, A. B. (1986). Technological opportunity and spillovers of r&d: Evidence from firms' patents, profits, and market value. *American Economic Review*, 76(5):984–1001.

Jaffe, A. B., Trajtenberg, M., and Henderson, R. (1993). Geographic localization of knowledge spillovers as evidenced by patent citations. *Quarterly Journal of Economics*, 108(3):577–598.

Katila, R. and Ahuja, G. (2002). Something old, something new: A longitudinal study of search behavior and new product introduction. *Academy of Management Journal*, 45(6):1183–1194.

Kneeland, M. K., Schilling, M. A., and Aharonson, B. S. (2020). Exploring uncharted territory: Knowledge search processes in the origination of outlier innovation. *Organization Science*, 31(3):535–557.

Kogut, B. and Zander, U. (1992). Knowledge of the firm, combinative capabilities, and the replication of technology. *Organization Science*, 3(3):383–397.

Kogut, B. and Zander, U. (1996). What firms do? coordination, identity, and learning. *Organization Science*, 7(5):502–518.

Lavie, D., Kang, J., and Rosenkopf, L. (2011). Balance within and across domains: The performance implications of exploration and exploitation in alliances. *Organization Science*, 22(6):1517–1538.

Levinthal, D. A. and March, J. G. (1993). The myopia of learning. *Strategic Management Journal*, 14(S2):95–112.

Levitt, B. and March, J. G. (1988). Organizational learning. *Annual Review of Sociology*, 14:319–338.

March, J. G. (1991). Exploration and exploitation in organizational learning. *Organization Science*, 2(1):71–87.

Nerkar, A. (2003). Old is gold? the value of temporal exploration in the creation of new knowledge. *Management Science*, 49(2):211–229.

Rivkin, J. W. and Siggelkow, N. (2003). Balancing search and stability: Interdependencies among elements of organizational design. *Management Science*, 49(3):290–311.

Rosenkopf, L. and Almeida, P. (2003). Overcoming local search through alliances and mobility. *Management Science*, 49(6):751–766.

Rosenkopf, L. and Nerkar, A. (2001). Beyond local search: Boundary-spanning, exploration, and impact in the optical disk industry. *Strategic Management Journal*, 22(4):287–306.

Schilling, M. A. and Phelps, C. C. (2007). Interfirm collaboration networks and knowledge creation: The impact of large-scale network structure on firm innovation. *Management Science*, 53(7):1113–1126.

Stuart, T. E. (2000). Interorganizational alliances and the performance of firms: A study of growth and innovation rates in a high-technology industry. *Strategic Management Journal*, 21(8):791–811.

Stuart, T. E. and Podolny, J. M. (1996). Local search and the evolution of technological capabilities. *Strategic Management Journal*, 17(S1):21–38.

Teece, D. J., Pisano, G., and Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal*, 18(7):509–533.

Trajtenberg, M. (1990). A penny for your quotes: Patent citations and the value of innovations. *RAND Journal of Economics*, 21(1):172–187.

Tripsas, M. and Gavetti, G. (2000). Capabilities, cognition, and inertia: Evidence from digital imaging. *Strategic Management Journal*, 21(10-11):1147–1161.